

Nathan MacLeod

macleod.nathanp@gmail.com

EXPERIENCE

AWS — Software Development Engineer

AUGUST 2023 - PRESENT

Design and implement significant new functionality for a worker scheduling management system written in Go. Ensure high performance and correctness.

Developing a FUSE based S3 backed Virtual File System written in C++. Improved performance by >80% by optimizing usage of locks.

UCSC — CSE 101 Tutor

JANUARY 2021- DECEMBER 2021

Responsible for explaining basic data structures and algorithms to students in, C / C++.

EDUCATION

UC Santa Cruz

SEPTEMBER 2019 - JUNE 2023

Computer Science B.S. & Mathematics B.A.

3.99 GPA

Summa Cum Laude, Highest Honors in Computer Science, Highest Honors in Mathematics, awarded Dean's honors all quarters.

SKILLS

Git, C / C++, Java, Python, Go, Multithreading, Optimization, Mathematics, Linear Algebra, Numerical Methods

PORTFOLIO

[GitHub](#) | [YouTube](#) | [Personal Site](#)

Some of the things I've made:

3d Physics Engine (C++)

- Multithreaded, Deterministic, and real-time. Supports multiple joint / constraint types. Uses a hybrid solver using PGS and LDL decomposition for holonomic constraints. [[Demo](#)]

Fourier Series

Visualization (C++)

- Generates a Fourier series that traces a drawn input. Also has an interactive demo for the Fourier Image Transform. [[Demo](#)]